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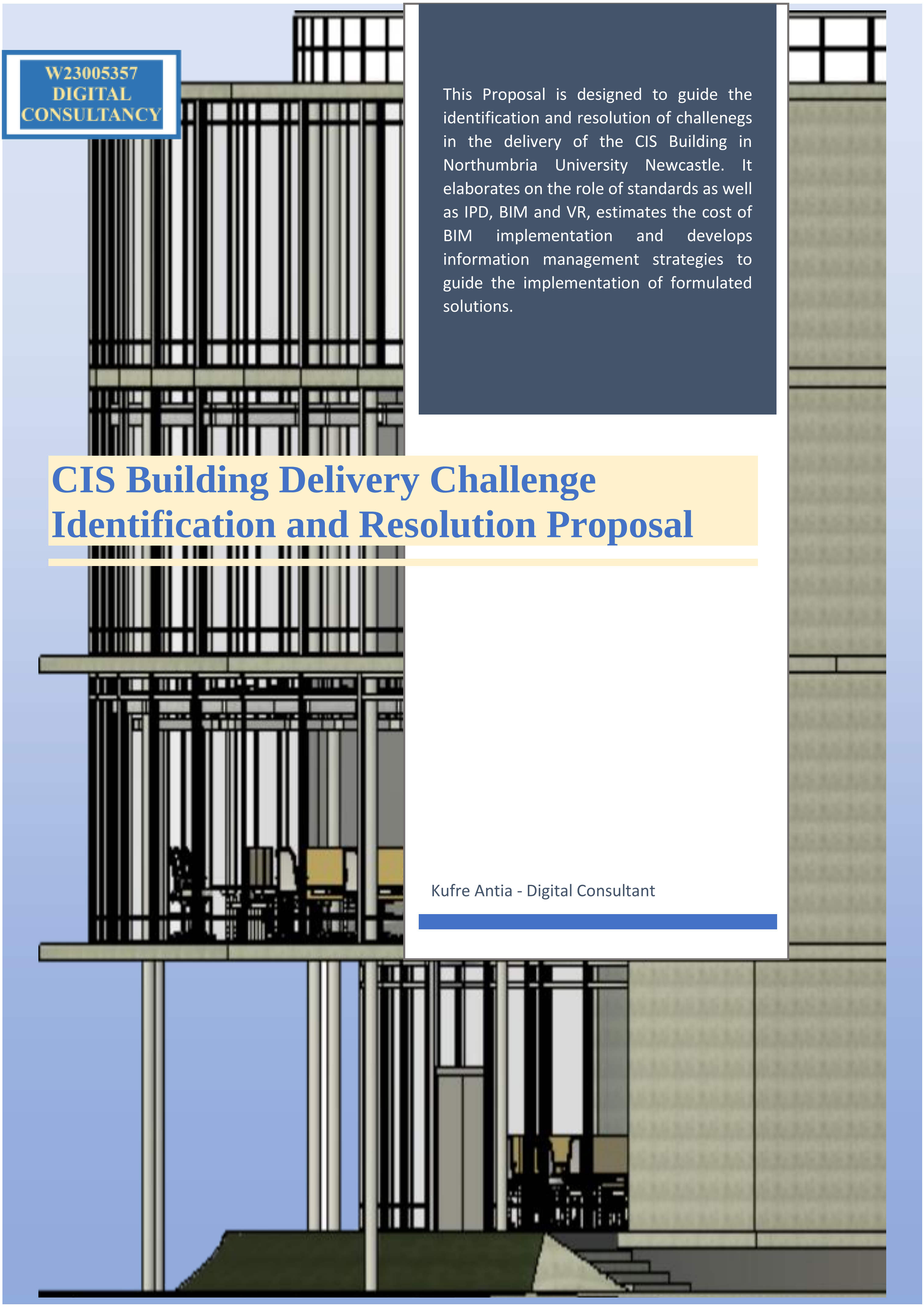

This Proposal is designed to guide the identification and resolution of challenegs in the delivery of the CIS Building in Northumbria University Newcastle. It elaborates on the role of standards as well as IPD, BIM and VR, estimates the cost of BIM implementation and develops information management strategies to guide the implementation of formulated solutions.

# CIS Building Delivery Challenge Identification and Resolution Proposal

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# 1.0 EXECUTIVE SUMMARY

The Computer Information Service (CIS) building is an educational infrastructure currently in the detailed design phase, designed to be a flexible and energy-efficient open-plan learning building. It encompasses lab spaces, research facilities and offices, spanning 520.78 square meters and stands at 16 meters in height, serving as an extension to the existing Elision building and aimed at modernizing it appearance and functionality.

The CIS building project adopts a Collaborative BIM approach, delivered at BIM stage 3 (Network-based Integration) through an Integrated project delivery (IPD) model facilitated by a Multi-Party Agreement (MPA). The BIModel is currently in a Use Shared State, hosted in a cloud-based Common Data Environment (CDE).

This proposal is structured to identify challenges and proffer solutions in the delivery of the project through the synergy between IPD, BIM and Virtual Reality (VR), analysing their roles. This synergetic approach enhances coordination, collaboration and communication during the design phase, thus reducing conflicts and ensuring stakeholder engagement across project stages.

Project stakeholders are individuals who can affect or consider themselves affected by the project. It includes the client as the Appointing Party (AP), the mono-disciplinary designers as the Lead Appointed Party (LAP) and third-party facet designers tasked with the design of specific components as the Appointed Party (TP-AP).

## *Challenge Identification and Resolution*

1. Functional, Aesthetical and Accessibility of CISc3 Circulation Corridor
   The integration of the architectural and structural model has revealed a conflict in the positioning of a structural column reducing the accessibility, functionality, and the aesthetical appeal of the layout of the CISc3 circulation corridor. The resolution of this challenge requires the modification of the corridor layout and requires efficient collaboration between the architect and the structural engineer, enabled by IPD.
2. Clash of Mechanical and Electrical Components
   The integration of the Mechanical and Electrical models has resulted in a clash of components on the first floor. This is addressed through clash detection, which is facilitated by BIM tools and enhanced through visualization in VR.
3. Modelling and Integration of Lift System
   The model of the mechanical Lift system is yet to be integrated into the model. As a highly energy-efficient building, a novel energy-efficient lift model is required. This necessitated the appointment of a TP-AP with the appropriate level of BIM capability and maturity to ensure proper information management and seamless flow of information and transfer from the Project information model to the Asset Information Model for facility management.

The proposed cost of BIM implementation for each mono-disciplinary team is estimated at £598,000.00 (See table 1). The wages of personnel is adapted from (JohnsonBIM, 2023)

# 2.0 IDENTIFIED CHALLENGES

To facilitate the identification and resolution of challenges in the delivery of the project, innovative project delivery methods such as IPD and standards such as ISO: 19650 as well as digital solutions within the auspices of BIM and VR are necessary to enhance collaboration, visualization, information management coordination and clash detection. The specific digital solutions, personnel and proposed implementation cost are detailed in Table 1.

| Type | Name | Specification | Description | Cost |
|---|---|---|---|---|
| Hardware | HP Victus Laptop | Processor 12th Gen Intel (R) Core i5-12450H | Processing Tool | £ 600.00 |
| | | Processor speed 2.00CHz | | |
| | | RAM 8GB | | |
| | | Graphic Memory 4.058GB | | |
| | | Graphics Card NVIDIA GeForce GTX | | |
| | Oculus Rift | H - FoV 87° V - FoV 88° | PC-powered VR headset | £ 600.00 |
| | | Pixel Density 12.04 PPD | | |
| | | Binocular Per-eye resolution 1080x1200 | | |
| | | Version 2024 | | |
| | | Version 2022 | | |
| | Oculus Rift Controller | | VR controller | |
| Software | Revit | | BIM Authoring tool | £ 3,000.00 |
| | Navisworks | | Integration tool | £ 3,000.00 |
| | BIMCollab | | | £ 800.00 |
| | Vrex | | A cloud-based VR Interface solution | TBD |
| Personnel | **Role** | **Description** | **Number** | **Yearly Salary** |
| | Digital Lead | Oversees BIM implementation and Digital Strategy | 1 | £ 80,000.00 |
| | BIM Manager | Leads BIM processes ensuring efficient Use of BIM for project Goals | 1 | £ 70,000.00 |
| | BIM Coordinator | Facilitates collaboration between disciplines using BIM Models | 1 | £ 60,000.00 |
| | BIM Technician | Creates and maintains BIM models under BIM coordinator's guidance | 4 | £ 180,000 |
| | Staff Training | Provides training for Project Personnel | Per Year | £ 100,000.00 |
| | Consultancy | Offers external expertise and guidance | Per Project Phase | £ 100,000.00 |
| | | | | **£ 463,000.00** |

*Figure 1: Table of Digital Technology, Personnel and Implementation Cost*

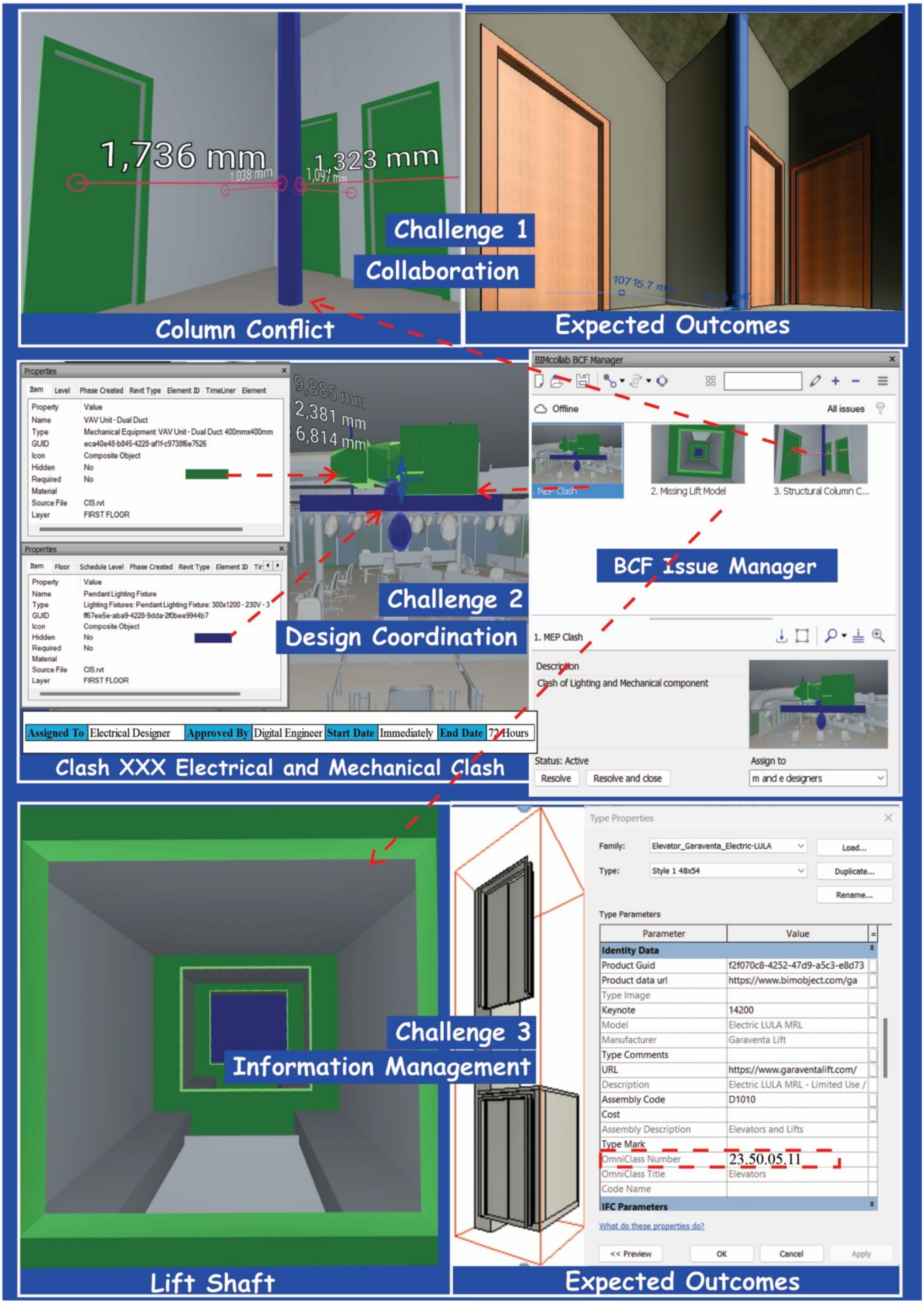


*Figure 2 Identified Challenges*

Figure 1 contains images of the identified challenges and expected outcomes after the implementation of the solutions recommended in proposal, including an Image of the BIM Collaborative Format File exported after the visualization of the identified challenges on VREX. The expected solution for challenge 2 is depicted in figure 2. It also contains component properties information highlighting OmniClass, a standard discussed in section 4 of this proposal.

### 2.1 Functional Aesthetical and Accessibility of CISc3 Circulation Corridor - Collaboration

Visualisation of the integrated model, enabled by VREX has resulted in the identification of discrepancies between the conceptual 2D drawings and delivered 3D model in the spatial arrangement, reducing the functionality and aesthetic appeal of the CIS1c3 circulation corridor. Also, the structural column between grids 5-6 and E-F challenges accessibility, and disrupts functionality and aesthetics (see Figure 1).

This challenge is a result of inadequate collaboration between relevant stakeholders, which could lead to delays, reworks and cost overrun in the construction phase. It can lead to accountability issues, where stakeholders may neglect responsibilities. This necessitates an integrated approach project delivery elaborated on in section 3.1 of this proposal.

### 2.2 Clash of Mechanical and Electrical Components – Design Coordination

The CIS building has extensive internal mechanical and electrical components on the first floor, hosted on the second-floor slab. Due to the complexity in positioning of these elements, the project requires thorough design validation of a variety of engineering disciplines.

In the ongoing coordination process within the CIS building project, the integration of MEP models and subsequent clash analysis and detection has revealed a hard clash in the positioning of an electrical pendant lighting fixture and a Variable Air Volume (VAV) Unit (see figure 1). The X,Y and Z coordinates of these two objects intersect necessitating proactive resolution measures. Clashes are termed as hard when two elements overlap in the same space (Abdalhameed & Naimi, 2023)

### 2.3 Modelling and Integration of Lift System – Information Management

The detailed design phase of an asset benefits from the integration of mono-disciplinary models, offering a holistic view of the project. The model of the mechanical lift system is yet to be incorporated into the integrated model, necessitating the appointment of a TP-AP to carry out this function in line with specific information management and exchange requirement to ensure the interoperability of produced data and information containers (see figure 1).

# 3.0 CHALLENGE RESOLUTION

## 3.1 Functional Aesthetical and Accessibility of CISc3 Circulation Corridor

The resolution of this challenge requires the adjustment of the layout of the CIS1c3 circulation corridor (see figure 1). This necessitates collaboration between the architectural and structural Lead Appointed Parties, involving the cost estimator to analyse the financial impact of this alteration. As an IPD project, the detailed design phase of the CIS building will benefit from the collaborative culture embedded in IPD, allowing for collaboration in resolving this challenge early on in the design phase of the project, thus mitigating the cost risk associated with the changes in the construction phase as represented on the Mcleamy curve.

### 3.1.1 IPD for Collaboration in Challenge Resolution

IPD is a project delivery method, which allows for adept collaboration and synergy between multiple parties. It as an approach to project delivery, which integrates systems, business structures, practices and people into a collaborative process that harnesses the expertise of all participants to optimize project outcomes, maximize efficiency and increase value through all phases of a project (AIA2007). Its basic principles encourage and motivate stakeholders to ensure that desirable project goals and outcomes are achieved. They add that the integration of technologies such as BIM and VR within the IPD process has offered a substantial increase in productivity, reduced field conflicts, and waste and enhanced project schedules. The specific role of IPD to the resolution of this challenge, based on its principles is detailed below.

### 3.1.2 Role of IPD

i. **Early Involvement of Key Stakeholders:** IPD encourages the involvement of key stakeholders early on, from the design phase of the project (Aapaoja, et al., 2013). They add that it reduces the probability of inefficient designs. This allows the design phase of the CIS building to benefit from early inputs of diverse expertise, facilitated through BIM and VR tools. The integration of mono-disciplinary architectural and structural design 3D models in Naviswork and visualization through VREX revealed the functional deficiencies of the CISc3 circulation corridor.

ii. **Shared Risk and Rewards:** Integral aspects of IPDs such as its approach towards project Risk and Rewards foster collaboration and enhances motivation within the project team. According to (Project Panagement Institute, 2017) Risk is the probability of an event and it subsequent consequences, representing the negative outcomes a project may encounter. Conversely rewards signify the positive returns or benefits accrued upon project completion. The Risk associated with project failure and the Rewards of success provide an incentive to task teams to ensure the successful resolution of this challenge.

iii. **Intensified Planning:** this is a key principle in IPD projects. It encourages increased efforts during the design phase allowing for iterative design and decision-making, which enables the assigned designers to continuously refine and improve the BIM model based on feedback on implemented changes, leading to increased quality of the BIModel. It also offers information on quantities.

iv. **Multi- Party Agreement (MPA):** This is an IPD contract model allowing for the integration of multiple parties in a single contractual agreement. According to (AIA2007) MPA enhances role interplay comprehension among parties due to the use of a single contract. The note that commitment of individual stakeholders is necessary for project success and as such, it requires trust and compensations are tied to the overall success of the project. In addition, they sight consensus decision making and role definition as key attributes of MPAs. The CIS building project is facilitated through a multi-party agreement (MPA) which specifies the rights, roles, liabilities and obligations of the primary project participants in a single contract, creating a temporary organisation to deliver the built asset.

### 3.1.3 Implementation Strategy

Having noted the influence of IPD in the ongoing resolution of this challenge, Figure 6 presents a sequential RACI Responsibility matrix for the implementation of this solution in line with the recommendation of clause 10.3 in ISO 19650-1:2018.

i. Initial Identification and Visualization using VR
ii. Collaborative Design Adjustment and Cost Analysis enabled by BIM
iii. Approval, Implementation and documentation guided by ISO 19650-2:2018

## 3.2 Resolution of Mechanical and Electrical Component Clash

A clash is defined as a conflict in the position of elements in a BIModel due to errors and occurs when components overlap in part or fully, sharing the same coordinates (Akhmetzhanova, et al., 2022). The pendant lighting component and VAV unit in grid D-E, 8-7 intersect. This poses a challenge to constructability and necessitates resolution by repositioning these elements for successful project delivery.

Clash detection is one of the most applied uses of BIM in the construction industry (Eadie, et al., 2020)It can be defined as an iterative process where conflicts in a BIM model are periodically identified, evaluated, classified and resolved; leading to a model with limited or acceptable levels of clashes (Wang, 2014). The resolution of this clash is achieved using Clash Detective in Navisworks, Visualization in VREX and collaboration facilitated through IPD.

### 3.2.1 Scoping the Role of BIM and VR

i. **Mono-Disciplinary Model Integration:** BIM tools such as Navisworks allow for the integration of mono-disciplinary electrical and mechanical models into an integrated model. This offers a holistic simulation, analysis and visualization of the BIModel, enhancing the detection of positional conflicts.

ii. **Clash Detection:** BIM enables comprehensive clash detection across building systems in an integrated model. The utilization of Navisworks clash detection tools enabled the identification and analysis of clashes between elements. Navisworks is an efficient BIM tool in clash detection and interference checks among model elements and can be used to check facets of the whole in terms of subsets or the entire composite model. It facilitates the recognition of clashes based on set rules, report generation, clash tracing, status and management (Kermanshahi, et al., 2020)

iii. **Immersive Visualization:** VR technology offers immersive visualization capabilities that allow stakeholders to experience MEP designs in a virtual environment. It enhances the understanding of spatial relationships thus facilitating better decision-making and communication regarding clash resolution.

iv. **Iterative Design Optimization:** BIM supports iterative design optimization, allowing for rapid prototyping and evaluation of alternative design solutions. The collaborative environment mandated by IPDs enhances it and benefits from the early involvement of stakeholders during the design phase of a built asset. This iterative approach enables the exploration of various resolution strategies, refines designs and validates solutions before implementation, reducing the risk of errors and rework.

### 3.2.2 Implementation

Having noted the influence of BIM and VR in the ongoing resolution of this challenge, Figure 6 presents a sequential RACI Responsibility matrix for the implementation of this solution in line with the recommendation of clause 10.3 in ISO 19650-1:2018.

**a. Clash analysis and identification**

i. Run Clash Detection in Navisworks and identify clashes

**b. Stakeholder collaboration**

i. **Assemble stakeholders**: Assemble relevant stakeholders to discuss the clash. The incentives associated with the shared risk and rewards from the MPA of the IPD will encourage attendance and effective collaboration

ii. **Visualization:** Utilize VREX to illustrate the clash and facilitate collaborative decision-making

iii. **Export Issue in BIM Collaborative Format (BCF) for Navisworks:** VREX enhance collaboration through its issue manager tools which allows for the exportation of issues in BCF for coordination on Navisworks.

iv. **BIMcollab**

c. **Resolution Strategy development:** Develop a comprehensive resolution strategy that addresses the hard clash by assigning responsibilities documented in a Responsibility Matrix. **Iterative design Implementation, decision-making and Validation:** Consider alternative positioning of these clashing elements by implementing the resolution strategy in coordination with relevant stakeholders (see figure 2). Validate the effectiveness of the resolution through a secondary clash detection and verification of MEP system compatibility and visualize the result in VREX by exporting an updated BCF file from Naviswork and importing it to VREX.

d. **Model Update**

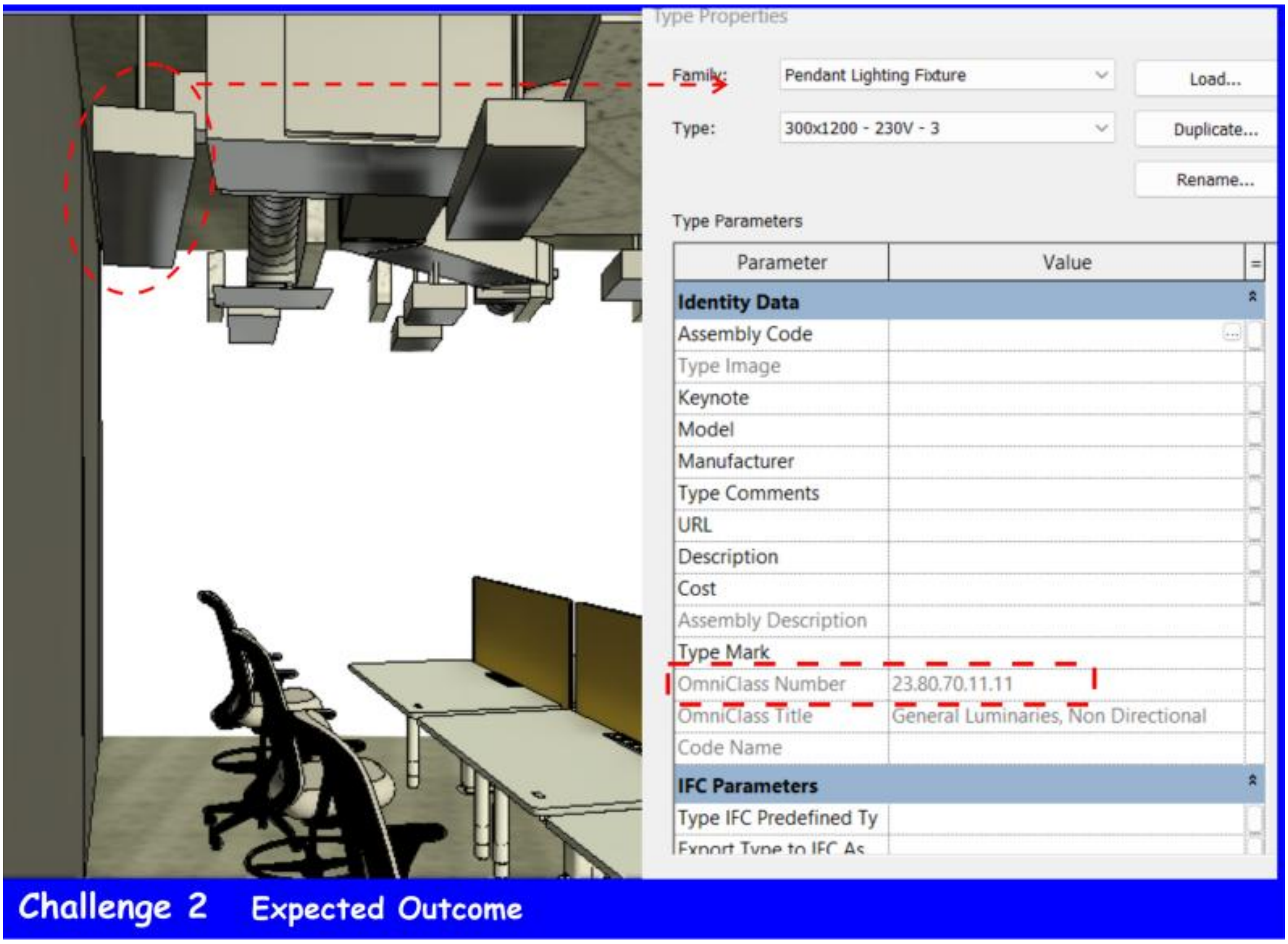


*Figure 3: Expected Outcome of Mechanical and Electrical Clash Resolution*

## 3.3 Information Management for the Modelling and Integration of Lift System

The resolution of this challenge requires the appointment of a third-party Appointed Party (TP-AP) by the mechanical Lead Appointed Party (M-LAP) for the design and modelling of a mechanical lift system in accordance with the specified Exchange Information Requirement (EIR). Collaboration is an integral aspect of BIM and relies on adept information exchange, necessitating scrutiny of platforms, data environments, formats, processes and policies to maximize efficiency and minimize errors (Gercek, 2016). This information exchange occurs on various platforms and as such requires that the information exchanged is interoperable across these platforms. Interoperability is defined as the ability to exchange data between applications, ensuring smooth workflows (Sacks, et al., 2018). According to (Laakso, 2012) Industry Foundation Class (IFC) is a non-proprietary open standard, which enables interoperability between BIM tools. Therefore, it is a prerequisite that the TP-AP has the capability and maturity for consistency in the development of the Lift model in line with the EIR.

### 3.3.1 Exchange Information Requirement (EIR)

The EIR is a crucial document that outlines the specific methods formats and procedures for information sharing between project stakeholders. It contains specifications regarding the what when how and for whom information is to be produced (International Organization for Standardization, 2018). The EIR outlines the technical, commercial and managerial aspects of the project's information production, with the technical aspect specifying information needed to address the Project Information Requirement (PIR) (International Organization for Standardization, 2018).

### 3.3.2 BIM Capability and Maturity Assessment

BIM capability refers to the minimum technical skills and functionalities an organisation possesses to deliver and benefit from model-based deliverables, reflecting the organisation's stage of BIM adoption, which ranges from Pre-BIM to Post-BIM, while BIM Maturity represents the quality, repeatability and predictability of an organisation's BIM process within each BIM capability stage and is measured through BIM Maturity Index (Succar, 2010). He adds that it is separated by BIM capability steps and measured through BIM capability stages. BIM capability and maturity access Technology (Software, Hardware and Network), process (Resources, Activities, Products, Services, Leadership and Management) and Policy (Preparatory, Regulatory and Contractual) and refers to the key performance milestones that can be achieved by an organisation as the adopt BIM tools, workflows and protocols. BIM maturity stages include

### 3.3.3 Responsibility Matrix

A responsibility matrix is a tool that clearly defines the roles and responsibilities of various stakeholders involved in a project. It is defined as a chart that describes the participation of various functions in task completion (International Organization for Standardization, 2018). It is mandated in clause 5.4.2 of iso19650-2:2018 for the appointment of the TP-AP.

The RACI matrix type of responsibility matrix that outlines the task on the vertical column and the stakeholders on the horizontal with R, A, C, I representing Responsible, Consulted, Accountable and Informed assigned to each stakeholder based on the required contribution to task completion. Figure 6 is a responsibility matrix designed to guide the implementation of the solutions to all the challenges in this proposal.

### 3.3.4 Selecting a Qualified Third-Party Appointed Party and Implementation

Selecting a qualified TP-AP for the design, modelling and integration of the mechanical lift system is crucial for project success. BIM Capability Assessment and BIM Maturity Matrix offer valuable evaluation tools for the M-LAP to evaluate the TP-AP's BIM experience and overall suitability. It gives insight into the organisation's BIM capability (skills and functionalities) and Maturity (consistency and predictability), which will inform decision-making for the appointment of a suitable TP-AP with the necessary expertise to deliver the Lift information model within established standards and information requirements, thus ensuring interoperability. A BIM Capability and Maturity Assessment form adapted from both (Arup, 2015) and (BIMe Initiative, n.d.) together with an ISO 19650-2:2018 guided EIR as mandated by clause 5.3.4 and clause 5.4.3 of ISO 19650-2:2018 been designed for this purpose (see figure 5 and 4). Also, in line with clause 5.4.2 a responsibility matrix has been created for the implementation of this process (see figure 6)

# EXCHANGE INFORMATION REQUIREMENT (EIR)

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Standard ISO 19650-2:2018 Clause 5.4.3

## Project Information

| Field | Name | Field | Name |
|---|---|---|---|
| Project Code | | BIM Manager | |
| Client/Owner | | M-LAP | |
| Project Manager | | TP-AP | |

Ref

**Objectives**

| |
|---|
| **Design**: Achieve high-level integration and coordination of the lift system model with other building systems. |
| **Construction**: Ensure the lift system model supports construction planning, scheduling, and clash detection. |
| **Operation**: Facilitate seamless transition of information to the AIM for facility management. |

## Information Requirements

a
b
c

**Design Phase**

| Requirement | Details |
|---|---|
| Model Development | Develop detailed 3D BIM models of the lift system using specified software. |
| Level of Detail (LOD) | Minimum LOD 300 during design development, progressing to LOD 400 |
| File Format | IFC (Industry Foundation Classes) and RVT including OminClass |
| Coordination and Clash Detection | Perform clash detection and provide clash reports using specified tools |
| Data Requirements | Include all necessary data attributes |

**Construction Phase**

| Requirement | Details |
|---|---|
| Construction Sequencing (4D BIM) | Integrate the lift system model with the construction schedule. |
| Cost Estimation (5D BIM) | Provide cost estimates based on the model. |
| Progress Monitoring | Update the model regularly to reflect construction progress. |
| Clash Detection Updates | Continue clash detection and resolution throughout construction. |

**Operation Phase**

| Requirement | Details |
|---|---|
| Asset Information Model (AIM) | Develop and deliver the AIM for the lift system, including operational data. |
| Data Handover Format | Deliver data in COBie, and read-only IFC format and native BIM format. |
| Integration with FM Systems | Ensure the lift system model integrates with the facility management systems |
| Documentation | Provide all relevant documentation, including manuals, warranties, and as-built drawings. |

| Role | Responsibilities |
|---|---|
| Client/Owner | Define overall project requirements and approve key deliverables. |
| Project Manager | Oversee project execution and ensure compliance with EIR. |
| BIM Manager | Manage BIM processes and coordinate between parties. |
| M-LAP | Ensure integration of the lift system model and manage TP-AP. |
| TP-LAP | Develop and deliver the lift system model as per EIR requirements. |
| Design Team | Support TP-AP in model development and integration. |
| Quality Assurance (QA) | Conduct quality checks and ensure compliance with BIM standards. |
| Facility Manager | Ensure the AIM meets operational requirements and integrates with FM systems. |

d

**Information Delivery Schdule**

| Phase | Milestone | Deliverables | Due Date |
|---|---|---|---|
| Design Phase | 30% Design Completion | Preliminary lift model | |
| | 60% Design Completion | Intermediate lift model | |
| | 100% Design Completion | Final lift system model | |
| Construction Phase | Start of Construction | Updated lift system model | |
| | Mid-Construction | Progress updates | |
| | End of Construction | As-built lift system model | |
| Operation Phase | Handover | Asset Information Model | |

e

| Aspect | Details |
|---|---|
| Collaboration Platform | Use a Common Data Environment (CDE) for all BIM data exchanges |
| Communication Protocols | Establish weekly coordination meetings for reporting and protocol |
| Issue Resolution | Implement a process for identifying, reporting, and resolving issues |

| Role | Name | Signature | Date |
|---|---|---|---|
| Client/Owner | | | |
| Project Manager | | | |
| BIM Manager | | | |
| M-LAP | | | |
| TP-LAP | | | |

*Figure 4: Exchange Information Requirement as per ISO 19650-2:2018 Clause 5.4.3*

# BIM Capability and Maturity Assessment

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| BIM Capability Set | | Initial (0-10) | Defined (10-20) | Managed (20-30) | Integrated (30-40) | Optimized (40-50) | Target Level | Current Level |
|---|---|---|---|---|---|---|---|---|
| Technology | Software | Usage unmonitored, mainly for 2D. | Unified usage for 2D and 3D. | Managed selection, 3D deliverables. | Strategic deployment, synchronized with processes. | Continuously optimized for productivity. | 35 | |
| | Hardware | Inadequate specifications, treated as cost items. | Suitable specs defined and budgeted. | Documented and well-targeted investment. | Treated as enablers, integrated with strategies. | Continuously tested and upgraded. | 35 | |
| | Network | Ad-hoc solutions, low bandwidth. | Defined solutions, basic connections. | Well-managed knowledge sharing. | Seamless real-time sharing, project-specific networks. | Continuously assessed and optimized. | 35 | |
| Process | Resources | Unconducive work environment, informal knowledge sharing. | Defined workplace tools, documented knowledge. | Controlled environment, adequately stored knowledge. | Integrated into performance strategies, accessible knowledge. | Constantly reviewed for productivity and satisfaction. | 35 | |
| | Activities & Workflows | Undefined processes, ambiguous roles. | Informally defined roles, independent planning. | Increased cooperation, consistent targets. | Integrated BIM roles, predictable productivity. | Continuously upgraded competency targets. | 35 | |
| | Products & Services | Inconsistent model detail. | Object breakdown statement available. | Adopted product/service specifications. | Differentiated per specifications. | Continuously evaluated and improved. | 35 | |
| | Leadership & Management | Varied visions, no guiding strategy. | Common vision, basic strategy. | Communicated vision, detailed action plans. | Integrated into all channels, competitive advantage. | Internalized vision, continuous realignment. | 35 | |
| Policy | Preparatory | Little to no training. | Defined training requirements. | Managed training to objectives. | Integrated into strategies, role-based training. | Continuously evaluated and improved. | 35 | |
| | Regulatory | Non-existent policies. | Basic compliance defined. | Monitored compliance. | Integrated compliance. | Proactively enhancing policies. | 35 | |
| | Contractual | No BIM-specific contracts. | BIM included in basic contracts. | Detailed BIM requirements. | Integrated into all contracts. | Continuously refined and optimized. | 35 | |

**Scoring Instructions:**

1. Assess each category by selecting the appropriate description that best fits the TP-AP's current practices.
2. Assign a score within the range specified for each level (e.g., if the TP-AP's software usage is managed with 3D deliverables, assign a score between 20-30).
3. Record the score in the "Current Level" column.
4. The percentage score is the sum of "Current Level" values divided by 500 and multiplied by 100

**Interpretation:**

**0% - 20%** - Initial BIM maturity

**20% - 40%** - Defined BIM maturity

**40% - 60%** Managed BIM maturity

**60% - 80%** - Integrated BIM maturity

**80% - 100%** - Optimized BIM maturity

**Calculation**

$\frac{\text{Total Score}}{500} \times 100$

**Target Level = 70%**

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*Figure 5: BIM Capacity and Maturity Assessment Form*

# IMPLEMENTATION RESPONSIBILITY MATRIX (RACI)

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## Challenge 1

| SQ | Task/Activity | BM | AL | SL | MC | CE | PM | QA | AP |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Initial Identification and Visualization | A | R | I | C | I | I | I | I |
| 2 | Collaborative Design Adjustment | A | R | R | C | C | I | I | I |
| 3 | Clash Detection and Report Generation | R | R | R | R | I | I | C | I |
| 4 | Clash Resolution Meeting | A | R | R | C | C | R | C | I |
| 5 | Update Models Based on Resolution | I | R | R | R | I | I | C | I |
| 6 | Structural Analysis | I | I | R | C | I | I | C | I |
| 7 | Cost and Feasibility Analysis | I | I | I | I | A | I | C | I |
| 8 | Quality Check of Updated Models | A | I | R | R | I | I | R | I |
| 9 | Final Model Integration | A | I | I | R | I | I | C | I |
| 10 | Client/Stakeholder Review and Approval | I | I | I | I | I | A | I | R |
| 11 | Documentation of Changes | A | R | R | R | I | I | C | I |
| 12 | Archiving and Model Handover | A | I | I | R | I | I | C | I |
| | | | | | | | | | |

## Challenge 2

| SQ | Task/Activity | BM | M-LAP | E-LAP | MC | PM | QA | DT | AP |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Detection of Clash | A | R | R | R | I | I | I | I |
| 2 | Clash Analysis and Report Generation | A | R | R | R | I | C | I | I |
| 3 | Clash Resolution Meeting | A | R | R | C | R | C | C | I |
| 4 | VR Visualization for Clash Review | A | C | C | R | I | C | R | I |
| 5 | Implementation of Clash Resolution | I | R | R | C | I | C | R | I |
| 6 | Model Update Based on Resolution | I | R | R | R | I | C | R | I |
| 7 | Quality Check of Updated Models | A | I | I | R | I | R | C | I |
| 8 | Final Model Integration | A | I | I | R | I | C | I | I |
| 9 | Documentation of Changes | A | R | R | R | I | C | C | I |
| 10 | Client/Stakeholder Review and Approval | I | I | I | I | A | I | I | R |
| 11 | Archiving and Model Handover | A | I | I | R | I | C | I | I |
| | | | | | | | | | |

## Challenge 3

| SQ | Task/Activity | BM | M-LAP | TP-AP | MC | PC | QA | DT | AP |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Selection of TP-AP | I | A | R | I | C | C | I | I |
| 2 | Capability and Maturity Assessment of TP-AP | A | R | C | C | I | C | I | I |
| 3 | Define BIM Requirements for Lift System | A | R | C | C | I | C | I | I |
| 4 | Contractual Agreement with TP-AP | I | A | R | I | C | I | I | I |
| 5 | Develop Initial Lift System Model | I | I | R | C | I | C | C | I |
| 6 | Integration of Lift System Model into Project BIM | A | R | R | R | I | C | C | I |
| 7 | Conduct Clash Detection Analysis | A | R | R | R | I | C | C | I |
| 8 | Resolve Clashes | A | R | R | R | I | C | C | I |
| 9 | Update Models Based on Clash Resolution | I | R | R | R | I | C | C | I |
| 10 | Perform Quality Checks on Updated Models | A | I | R | R | I | R | C | I |
| 11 | Final Integration of Lift System Model | A | R | R | R | I | C | C | I |
| 12 | Document Changes and Updates | A | R | R | R | I | C | C | I |
| 13 | Client/Stakeholder Review and Approval | I | I | I | I | A | I | I | R |
| 14 | Archiving and Handover of Final Models | A | I | R | C | I | C | I | I |
| | | | | | | | | | |

| LEGEND | |
|---|---|
| AP | Appointing Party |
| AL | Architectural Lead |
| BM | BIM Manager |
| CE | Cost Estimator |
| DT | Design Team |
| E-LAP | Electrical Lead |
| MC | Model Coordinator |
| M-LAP | Mechanical Lead |
| PM | Project Manager |
| QA | Quality Assurance |
| SL | Structural Lead |
| TP-AP | Third-Party Appointed Party |

| LEGEND | |
|---|---|
| R | Responsible |
| A | Accountable |
| C | Consulted |
| I | Informed |

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*Figure 6: Implementation Responsibility Matrix (RACI)*

# 4.0 Standards in BIM Project Delivery

Standardization is essential to efficient information leveraging on a BIM Project, as information must be universally understood and accessible (Weygant, 2011). Working on IPD requires collaboration, this necessitates proactive measures to ensure that interoperable information and data can be shared among project teams and archived for future reference seamlessly. Interoperability of data is an integral aspect of collaborative BIM.

## 4.1 ISO 19650 Part 1 and 2

ISO 19650 series established unified international framework for collaborative information management within the construction industry, focused on ensuring timely and pertinent information access for relevant staveholders (12d' Synergy, n.d.). ISO 19650-2:2018 has outlined information management processes, which serve as a guide during the delivery phase of an asset. The table below highlights some of the relevant clauses adopted in the resolution of these challenges.

***ISO 19650-1:2018 Clause 10.3***

This mandate the development of a responsibility matrix as part information delivery planning process as executed in Figure 6

***ISO 19650-2:2018 Clause 5.4.3***

This mandates the establishment of exchange information requirements for each appointed party by the LAP. It defines information requirements and level of information need as well as acceptance criteria, dates and information need of the appointed party (See Figure 4).

***ISO 19650-2:2018 Clause 5.3.4***

This clause mandates the assessment of the BIM capability of the appointed party to deliver information in accordance with the Lead Appointed party's EIR defined in clause 5.4.3 and discussed in section 3.3, and developed in Figure 5 of this proposal.

## 4.2 OmniClass

Omniclass structures BIM data to ensure consistency and accuracy in BIModels, facilitating better collaboration and communication among project stakeholders. It is a comprehensive classification system used primarily in the construction industry comprising a list of tables which enhances the organisation of information contained within a model and allows for cross-referencing (Weygant, 2011). In the delivery of the CIS building, Omniclass plays an integral role in the classification of component such as Lighting elements (See figure 3 ) and the Mechanical Lift (see figure 1).

The classification of the Lift incorporated into the model as shown in Figure 1 is 23.50.05.11 and is broken down as follows

- Primary Level – 23 – This corresponds to the product table and allows for specific building materials and products to be classified within a project.

- Secondary Level – 50 – This represents a division within the product table, directly related to conveying systems.
- Tertiary Level – 05 – This further breaks down the division aiming to reduce it to the simplest level (Weygant, 2011). It relates to vertical transportation.
- Quaternary Level – 11 – This level offers even more specificity, directly relating to Lifts.

The quaternary level can be further reduced with a fifth level “11” representing Electrical Traction Lifts a sixth level “14” passenger electric traction lifts (OmniClass 23 Products List, 2017)

This classification ensures that each element in the BIModel is accurately identified and organised accordingly. It ensures interoperability and helps maintain consistency across platforms and among stakeholders. This improves database integration and aids in detailed cost analysis and estimations, resulting in accurate budgets, procurement and cost control.

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